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APPG on
Diversity and
Inclusion in STEM

# Equality, Diversity and Inclusion strategies in STEM - briefing

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Secretariat:

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In May 2024, the All-Party Parliamentary Group (APPG) on Diversity and Inclusion in STEM launched a new project on Equality, Diversity, and Inclusion (EDI) strategies in STEM to explore the most effective approaches to progressing EDI and improving representation in the STEM workforce. As part of the project, we held a series of interviews with EDI leads and Human Resources (HR) departments at various STEM organisations about their EDI strategies. In this briefing, we summarise our main findings from this research, and identify areas where action is needed.

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*The views expressed in this report are those of the All-Party Parliamentary Group on Diversity and Inclusion in STEM. The report does not necessarily represent the views of any of the sponsors, contributors and reviewers, unless stated. This report was compiled by the British Science Association who provide the Secretariat to the APPG. The information in this briefing, including reference to policies and Government status on issues, is correct at the time of writing (January-March 2025).*

*Drafts of this briefing were sent for critical review to a group of reviewers. In addition, the draft was reviewed by our sponsors and Members of the APPG.*

# Equality, Diversity and Inclusion Strategies in STEM - briefing

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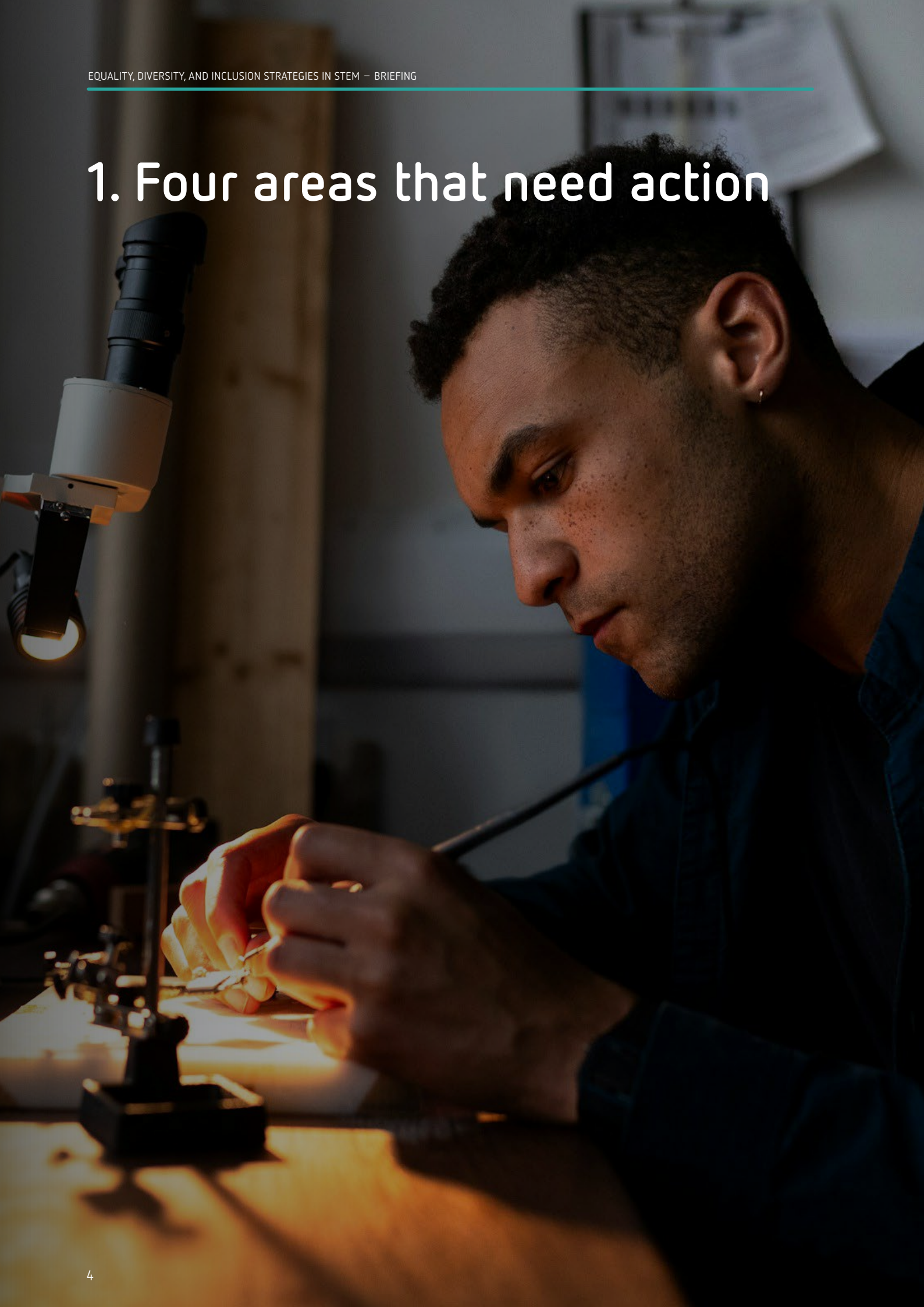
# 1. Four areas that need action



## 1. A stronger case for EDI as central to growth and performance:

EDI is linked to stronger commercial performance, improved talent attraction and retention, better customer engagement, increased investment, and reduced costs associated with legal disputes. Diverse teams are more innovative, produce higher-impact work, and are better at tackling complex problems[1], which is crucial for the STEM sector.

A more diverse sector with the skills the UK needs for growth requires the collaboration of STEM leaders in companies, universities, professional bodies and associations, and Government. We recognise the competing pressures and priorities facing STEM organisations, and the continued need to emphasise how diversity and inclusion supports growth, business objectives, and better outcomes.

a. We call upon all STEM leaders to recognise the essential role that a diverse workforce and an inclusive culture plays in business growth and performance, and the need for EDI strategies to go hand in hand with overall business strategies. To demonstrate this commitment, leaders should establish measurable EDI targets and actions with clear accountability mechanisms, and report progress annually to stakeholders alongside financial and operational performance metrics (for large STEM companies this could be broad quantitative targets; while for smaller organisations targets could be to embed EDI training for all staff or to introduce a recruitment practice that draws on best practice in EDI).

b. We recommend the Department for Science, Innovation, and Technology (DSIT) and the Women and Equalities Unit identify practical, immediate steps that can drive progress within UK STEM organisations, for example:

   i. Compliance: Sector-wide engagement scheme to ensure that all STEM organisations are meeting the legal duties introduced in 2024 as amendments to the Equality Act 2010. These amendments include: including measures to protect rights during pregnancy, maternity and as a breastfeeding parent; creating transparency for equal pay for disabled employees and those from ethnic minorities; and wider responsibilities of employers to prevent sexual harassment.

   ii. Procurement: An assessment of how the Social Values Framework and Public Sector Equality Duty can be better applied to improve EDI within STEM organisations granted Government contracts.

[1] Page, S. E. (2007). The difference: How the power of diversity creates better groups, firms, schools, and societies. Princeton University Press.

## 2. Improving data accessibility:

Whilst there is high-level data on a national level, it is not granular or accessible enough to support decision-making. Additionally, STEM organisations are struggling with intersectional analysis of data, and declaration rates, particularly relating to disability and socioeconomic status.

a. We recommend further investments in data-dashboards such as the Department for Education's Jobs and Skills Dashboard, to extend the demographic data beyond gender, and to provide user-guides to STEM industries.

b. As a first step, this APPG will convene a discussion around data collection, insight, and analysis for diversity and inclusion in STEM, focusing on improving national-level workforce and education data. In a future programme of work, this APPG will consider how to support more widespread and trustworthy data collection on national and organisational levels, including better data hygiene to help organisations protect and interpret the data well.

## 3. Evaluating the impact of EDI interventions:

Whilst there is plenty of evidence to inform where EDI interventions are needed, it is rare that interventions are accompanied by outcome evaluations, particularly when concerned with workplace culture and retention. This is limiting the ability of the STEM sector to take evidence-driven approaches and learn from what is working. Additionally, sector-wide frameworks are important tools for employers, however, they can be time-intensive to complete, open to 'gamification' and vulnerable to changing external environments.

a. We recommend that STEM organisations carry out an equity impact assessment for key changes in the business or organisation, particularly around retention and workplace culture.

b. We recommend that DSIT collaborate with sector partners from the corporate sector, higher education, and key agencies to support the industry by fostering a financially sustainable ecosystem of diverse, purposeful EDI frameworks in STEM, and ensuring the approach to data collection is more consistent.

## 4. Understanding the challenges for start-ups, spin-outs, and scale-ups:

Start-ups and scale-ups are important for the UK's STEM ecosystem, but they have very different needs to large STEM companies.

a. We recommend a sector-wide engagement scheme on diversity and inclusion for start-ups, spin-outs and scale-ups, which takes learnings from initiatives like the Royal Academy of Engineering Culture+ tool.

b. We call upon large STEM organisations with the resources to develop EDI toolkits and training programmes to more proactively share with smaller organisations.

# 2. Context: EDI strategies and the STEM sector

# Context: EDI strategies and the STEM sector

## State of the sector

Diversity of thought and experience is critical to unlock creativity and innovation and enhance profitability, which means that a lack of representation prevents the STEM sector from reaching its full potential. Growing the STEM pipeline and making the most of everyone's skills is therefore not only the right and fair thing to do, but it also makes sense economically.

It is widely recognised that STEM skills are vital for the future of UK industries. Driven by technological advancements and sustainable development, our economy is likely to be dominated by STEM industries, with engineering and computer sciences providing some of the highest paid jobs[2]. But for STEM industries to stimulate economic growth, we need more people with STEM skills.

Many sectors and employers are facing 'skills gaps', finding it difficult to recruit workers for more technically demanding jobs. Extra costs associated with recruitment, temporary staffing, and training have been estimated to cost the UK economy £1.5 bn per year[3].

The shortage of STEM skills in the UK has been a concern for the past two decades, and the problem will likely be exacerbated in the future by the changing world of work. It is estimated that by 2030, 90% of the current workforce will need upskilling and retraining[4]. Additionally, many STEM employers operate worldwide, and if there is not a ready supply of STEM skills in the UK, there is a risk that the sector 'offshores' the most advanced roles and loses the potential for growth in the UK.

Tackling the skills gap, future-proofing the STEM workforce, and harnessing sectors such as AI, life sciences, and climate technologies, will lead to job creation, increased productivity, and wider economic growth.

But we know that opportunities to work in, learn about, and progress in STEM are not available to everyone. Our research from 2020[5] found that:

- Only 27% of the STEM workforce are women (compared to 52% of the wider workforce)
- 65% of the STEM workforce are White men
- Disabled people represent 11% of those employed in STEM (compared to 14% of the wider workforce)
- STEM workers are less likely to be aged 50 or older (28% compared to 33% of the wider workforce)

Research by the Royal Academy of Engineering[6] found that embracing EDI is linked to stronger commercial performance among engineering businesses. For example, companies with higher diversity have been found to generate 19% more revenue from new products and services and see 121% more patent citations.

[2] World Economic Forum. (2023). Gender equality in STEM can support a sustainable economy. Here's how

[3] STEM Learning. (2018). STEM Skills Indicator

[4] CBI. (2020). Learning for life: Funding a world-class adult education system

[5] APPG on Diversity & Inclusion in STEM. (2020). The State of the Sector: Diversity and representation in STEM industries in the UK

[6] Royal Academy of Engineering. (2024). The EDI Engine: Evidencing the business benefits of equality, diversity, and inclusion (EDI) in engineering

## *How can EDI strategies help?*

EDI widens the pool of STEM skills available to employers and helps to improve productivity and innovation. By putting in place an effective EDI strategy, organisations not only commit to good people management and the wellbeing of all their employees but also support their business objectives[7]. It is important to emphasise that being inclusive benefits all employees, not just people from groups underrepresented in STEM. Outlining plans for recruitment, retention, workplace culture, and evaluation helps organisations to add value to their overall business strategies and drive systemic change in the sector.

Developing and implementing a successful EDI strategic plan can be a challenge. Organisations need a good understanding of the demographic composition of their workforce, but many employers struggle with collecting data about their employees. Based on their industry-wide EDI Survey, the Investment Association[8] reports that limitations of existing HR systems (difficulties around collecting and analysing data points) and not being mandated by current regulations were the main challenges to data collection at the organisation level. Concerns over data privacy and lack of employee engagement have been found to be the two top challenges at the individual level.

[7] CIPD. (2024). Equality, diversity, and inclusion (EDI) in the workplace

[8] The Investment Association. (2023). Equity, diversity, and inclusion data survey

# 3. Summary of sector interviews

## Leadership

Strong and clear leadership was a key factor for the success of EDI strategies among interviewees. For interventions to work at scale, EDI policies need to have a buy-in from senior leaders and the board. Sometimes, EDI strategies originated from leadership, and other times from staff networks, but in all cases we heard about how crucial the support and commitment of leadership was to integrate EDI into the organisation.

Participants also stressed that the EDI strategy needs to go hand in hand with the overall business/university/growth strategy, and that significant change needs long-term investment. To hold leadership accountable, EDI needs to be treated similarly to finances, and checked, reviewed, and reported on regularly.

## Data about diversity characteristics

**Data as the biggest challenge** – Most participants said that accessing and analysing diversity characteristics data was the biggest challenge of developing their strategy. Generally, participants were happy with their data on gender representation in their workforce, and some have also described their data on ethnicity as "good".

When asked about missing data that would help the organisation with their EDI work, several participants mentioned better data on disability and socioeconomic status.

We have also heard that the ONS data doesn't provide enough granularity on the working population, and that better understanding of the UK workforce is needed to support benchmarking.

**Improving data about diversity characteristics and disclosure rates takes time.** This can be a real barrier for STEM organisations who are typically very data-driven. Senior leaders are often not willing to move forward and implement changes unless they see clear data to go with it.

On the other hand, employees want to see results quickly. They volunteer their time to make the organisation a more inclusive workplace and want to see the senior leadership committed to EDI too. But unpicking a culture takes time - there is a conflict between a need for speed and for introducing changes in a sustainable way.

One participant said:

*"You can't change the diversity of your workforce overnight – managing people's expectations about how long it's going to take was a challenge."*

**Intersectionality** – We've heard from participants about the importance of intersectionality and not working in silos on certain groups and protected characteristics to make changes.

One participant told us that when zooming in on a specific issue, the data needs to be looked at intersectionally, but when you work with relatively small numbers, individuals can quickly become identifiable. This means that it can be difficult to advocate for certain groups because you can't present the data.

**Too much data?** Several interviewees mentioned the problem of gathering too much data. While it can be tempting to consult everyone when developing a strategy, this is not always helpful. Fewer but better focused focus groups, and better data rather than more data, is usually a more productive approach.

Participants told us:

*"It's one thing to have data, but another thing to know what's going on"*

*"We're now moving to a stage where it's not about collecting more data - perhaps it's about collecting less data but using it better."*

**Protected characteristics** – In the UK, the legal framework for EDI work is built around the nine protected characteristics named in the Equality Act 2010, which all employers have a legal duty to abide by. But, there are many characteristics that are not included in the Equality Act 2010 (for example, to include socio-economic disadvantage, paternity and shared parental leave) and one participant noted that not everyone identifies themselves in the existing categories. They said:

*"It can be difficult to find a balance between collecting data on identity and background, but also not focusing on this too much and allowing people to come to work or study no matter what different boxes they might choose to tick on a page."*

**Data accessibility** – One participant pointed out that their organisation had a lot of data available, but it is not always available in a user-friendly format. There is a need for dashboards that allow users to use different filters and look at a specific issue. EDI experts need to work with data experts to make that happen.

**Data privacy and disclosure rates** – Several participants said that encouraging employees to disclose their data was ongoing work. Interviewees thought the possible reasons for low disclosure rates were that people did not see the task of filling out a staff survey and sharing their data as important – because it is not compulsory, it is never a priority on anyone's to-do list.

Some participants have said that staff concerns about data privacy is a big issue for them and that the problem might be prevalent at STEM organisations, particularly among computer scientists and those working in cybersecurity who have a good understanding of how the data might be incorrectly accessed or breached. One participant has said:

*"It's not that they don't trust the HR, they don't trust the hackers."*

Some of the organisations we spoke to said they run regular campaigns to encourage people to complete a staff survey and inform employees how sharing their data can help the organisation become a fairer, more inclusive, and more successful place to work.

**AI in STEM recruitment** - An emerging area related to data collection and use is the use of AI and digital tools in recruitment. The use and ethics of AI tools was not mentioned by interviewees.

## Benchmarking and impact

**Benchmarking data** – While it is useful to have this, organisations need to be careful about comparisons and the weight they put on it (for example, there are geographical effects that are not always evident in benchmarking data).

One participant told us that benchmarking data serves better as a reflective tool than as a clear way to see where progress is happening – you need to be sure what you are trying to achieve locally and within your system, not necessarily wanting to get to national averages. Participants have said:

*"People can get a bit caught up in benchmarking data. It's helpful, but you have to understand the system."*

*"Don't separate data from context, stories, and experiences – it's unhelpful."*

*"It's easy to copy others, but it means that sometimes we just replicate foolishness."*

**Diversity targets** – In general, people we spoke to had mixed feelings about standalone diversity targets. While some said that they can be helpful and useful as part of a system of measures, others have pointed out that they can be distracting or even drive wrong behaviours.

One participant noted that they found targets focused on reducing the gender pay gap more helpful for driving change than workforce diversity targets.

**Performance data is a challenge** – We have heard repeatedly that organisations often struggle with demonstrating the impact of their EDI work beyond a shift in the diversity of staff. Better access to tools and frameworks for tracking performance would be welcomed by the sector.

**Accreditations** – All organisations we spoke to had at least one accreditation (most often Disability Confident). People said they were generally useful and have helped to push the dial and bring awareness. However, participants also noted they can be very time-consuming and that their organisations would not have the capacity to apply for another one.

Interviewees working in academia stressed the importance of the Athena Swan Charter, which has helped to put EDI on the map for universities.

Several participants noted that very long and complex application process can be a burden for those responsible (often women), and their careers would suffer as a result.

**Tech Talent Charter** – A couple of participants said the Tech Talent Charter (TTC) had really helped them to benchmark themselves against other tech organisations, and they were not sure how to do this since the TTC community closed.

## Other topics

**Retention** – We heard relatively little detail about strategies targeting retention and workplace culture and how these important elements of EDI are measured.

**Staff networks** – Almost everyone talked about them and how important their input was in developing the strategy. It can be challenging to capture lived experience and make sure that this is reflected in the strategy – staff networks have a key role to play in this challenge.

A couple of participants cautioned against leadership depending on staff networks to lead EDI work – staff networks are often led by volunteers, and they should not be expected to transform organisational culture.

**Resourcing EDI** – When asked about the challenges of developing and implementing the strategy, participants often talked about the lack of or limited resources (e.g. no ringfenced staff resource). We heard that it would be useful to know what type of investment is most needed in EDI across the whole STEM sector (for example, what the balance should be between grassroots investment at school level, entry-level, middle management, and reskills/upskilling at senior level).

**School outreach** – A couple of participants spoke about the importance of school outreach to inspire the future STEM workforce. They said that while senior leadership saw the value of engaging with universities and future or recent graduates where there is clear return on investment, it was usually more difficult to convince them to engage with local schools or schools in areas with high populations of underrepresented groups. However, we know that perceptions of STEM start early, and some of our interviewees were investing in early-intervention, rather than older students. One participant said:

*"We're really passionate about changing the direction of people at primary school, because that's where the change needs to happen"*

**EDI as a term** – Equity, diversity, inclusion, and belonging are all different things, so some questioned the usefulness of 'EDI' as a term. We hear people say that it is easier to talk about diversity, representation, and the data that sits behind this – whereas having a conversation about inclusion and how to measure it is much more difficult. Because of the lack of confidence with measurement and qualitative data, it is often the case that success only gets measured around diversity.

### Equity or equality?

Equity and equality are two approaches used to try to create fairness. This briefing defines 'equity' as a needs-based approach to ensure people with different needs can be equally as successful, and 'equality' as treating everyone the same.

In our APPG work, we usually choose to use 'equity' as many studies state that treating everyone equally within the workplace and education tends to lead to unequal outcomes, as it assumes all workers begin in similar circumstances, underestimating the impacts of structural inequalities.

**EDI strategies at SMEs** – While bigger organisations often focus their EDI efforts on gender and ethnicity, for smaller organisations, the lack of people coming from lower socioeconomic backgrounds is often a challenge. People told us about the problem of nepotism in the industry – it is a common practice for, for example, directors to invite their children and relatives to run the business with them. As they scale-up this then makes it difficult for organisations to be inclusive.

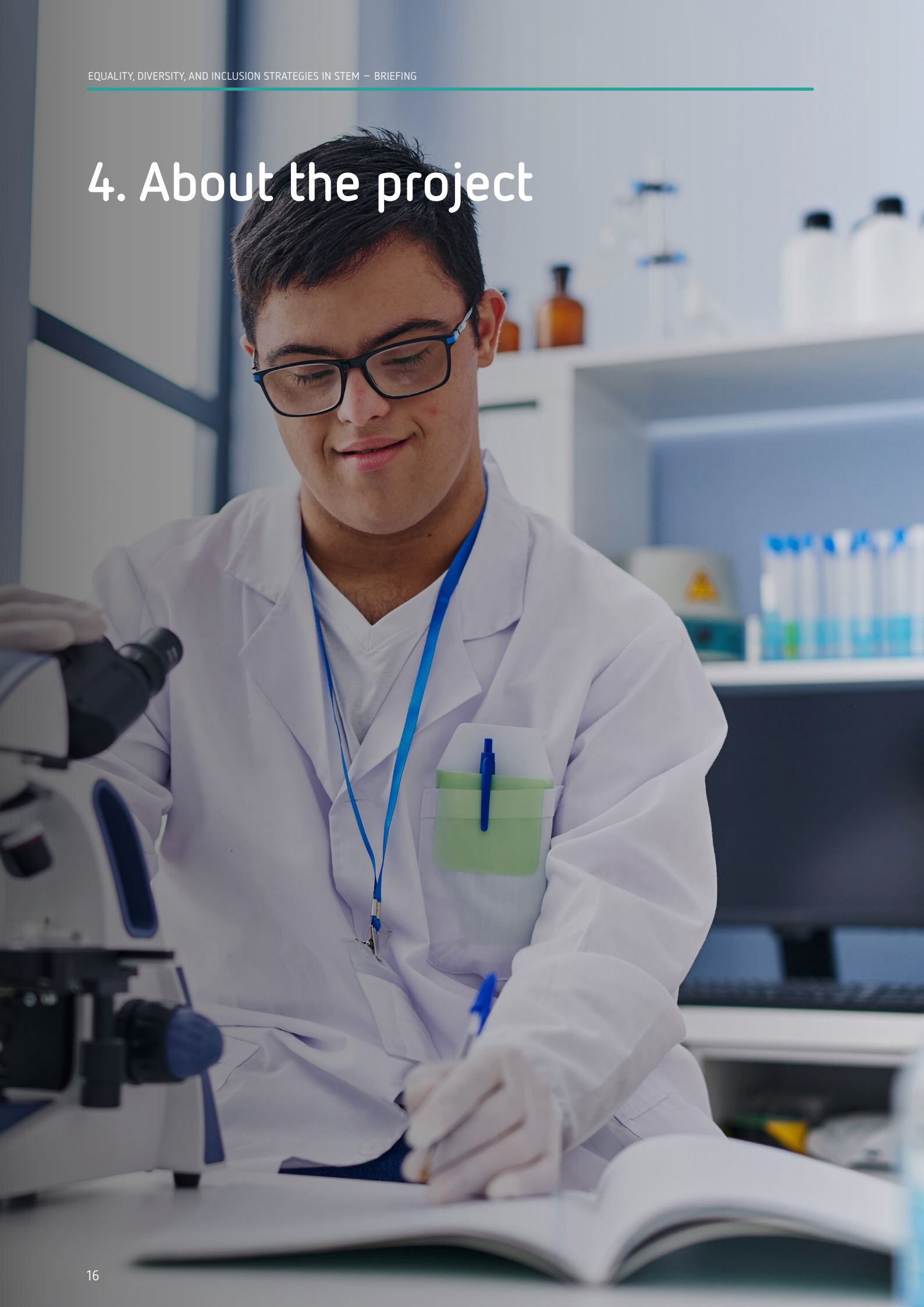
We also heard that there is not as much time and space to develop people, and it is a common practice to use referrals and go for the 'safest pair of hands' (e.g. one person said that biotech startups would usually recruit graduates directly from Oxford or Cambridge rather than running an open process).

It can also be difficult for smaller organisations to learn from large corporations. There are a lot of sector-specific networks that include both large and small businesses, but it is usually more useful to connect with businesses with a similar profile.

Another problem is that the smaller size of the staff team makes it more difficult to meet targets that benchmark against the wider workforce. For example, in an organisation with only 10 employees, if a woman leaves and her role is taken by a man, this will immediately make 10% difference to the company's diversity statistics.

**EDI strategies at universities** – Two participants spoke to us about their work on decolonising the curriculum as part of the EDI work with the student body. EDI data revealed awarding gaps between ethnic minority and White students, and that this was particularly significant in STEM subjects. When further investigated, they heard from students that there were issues with the teaching systems. We were told how important it had been to challenge colonial views in STEM curricula, for example, issuing guidelines for lecturers, and working with students on co-creating the curriculum.

# 4. About the project



In May 2024, the APPG on Diversity & Inclusion in STEM launched its new project investigating equality, diversity and inclusion (EDI) strategies in STEM.

The project has evolved from an idea submitted by Professor Nira Chamberlain, a mathematician and immediate past president of the Mathematical Association, to the open call held by the APPG in early 2024. Nira advocates for treating EDI as a pure science problem with an established baseline that organisations should use to establish targets. In his submission, he called for more research into EDI strategies and how they could better reflect the lived experience of groups underrepresented in STEM.

From the 15+ submissions received and following feedback from members of the APPG, the Secretariat chose the final concept – EDI strategies in STEM. The project investigates how EDI strategies are developed and the evidence used by STEM organisations to form their approaches to tackling EDI.

Collecting people data and setting ambitious diversity targets can help employers improve representation of people with protected characteristics in their workforce. However, we know that focusing on diversity alone isn't enough to address systemic barriers to equality and bring about meaningful change to individuals and communities.

To progress the EDI agenda and improve representation in the STEM sector, it is necessary to improve the quality, and broaden the types, of gathered data and ensure that its analysis considers the lived experience of groups underrepresented in STEM.

As it stands, evidence for such analysis is typically quantitative, and therefore does not capture any of the emotional and personal drawbacks of working in STEM.

In this project, the APPG explores what can be done to improve data reporting on representation in the STEM workforce and how organisations can evaluate the effectiveness of their EDI action plans. We will do that by:

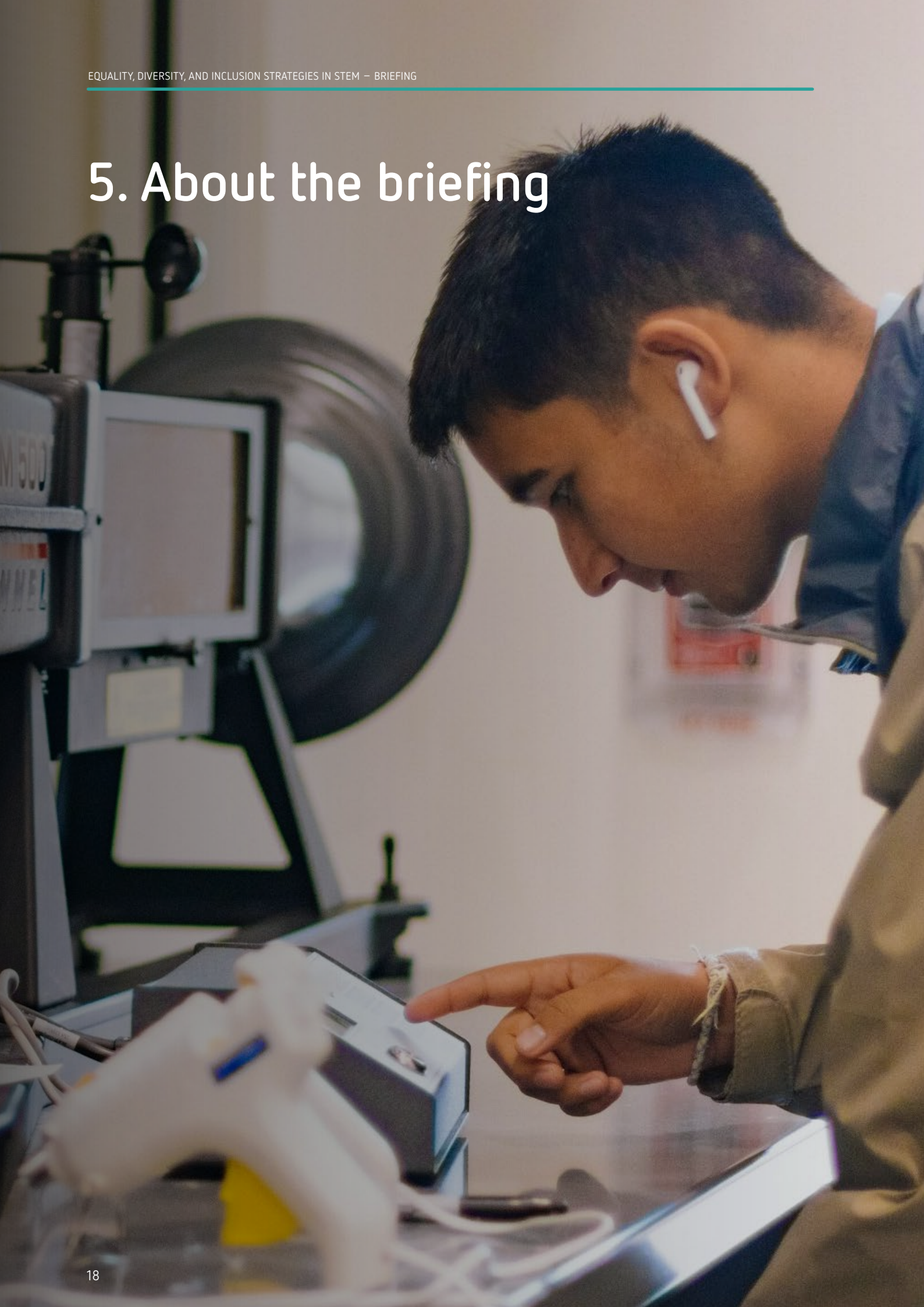
- Undertaking a series of interviews with EDI leads and HR departments at universities, research organisations, and STEM businesses to learn about the process of developing an EDI strategy and the evidence behind it,

- Hosting a parliamentary roundtable with key STEM organisations about EDI data collection, analysis and insight to discuss the common challenges that STEM organisations face with collecting EDI data and exchange best practice,

- Publishing a series of blogs by EDI experts and network leaders to outline their ideas on how EDI strategies can better reflect lived experience.

This briefing is the first publication of this project, and its aim is to explore the main themes which emerged from our interviews with STEM organisations about their EDI strategies.

# 5. About the briefing



To learn more about how EDI strategies are developed at STEM organisations, we held a series of online interviews with EDI leads and HR departments at universities, research organisations, and businesses between July and November 2024.

Each interview lasted approximately 40 minutes and followed a similar structure. We asked the participating organisations about the:

- process of creating the strategy (How was your EDI strategy created? Who was involved? What were the challenges?),
- evidence behind the strategy (What data is your strategy based on? Have you used any frameworks? What missing data would make a difference?),
- and impact of the strategy (How are benchmarks set? How is success measured? Do you have any accreditations? What positive change has the strategy helped to achieve?).

To develop a good understanding of the different challenges that STEM organisations face when working on their EDI action plans, our aim was to speak to a wide variety of stakeholders. When selecting organisations to invite to participate in the project, we considered: size and type of the organisation, area of expertise and activity, and location.

We approached 53 organisations from across the STEM sector, and secured interviews with 14 of them. It is important to acknowledge that organisations who agreed to speak to us are likely to be organisations who are some of the best resourced and most confident in their approaches to EDI. This means that many organisations in the sector might be less advanced in planning their EDI strategy and in need of more support to address underrepresentation in their workforce.

This briefing explores the main themes of these conversations, and aims to shed light on some of the challenges that STEM organisations face when developing and formalising their approaches to EDI. We also recommend steps and suggest actions that would support organisations in progressing their EDI agenda and becoming more inclusive workplaces.

We would like to thank the following organisations for participating in the process: The Alan Turing Institute, AtkinsRéalis, AVEVA, Barclays, Brighton Sussex Medical School, Hitachi Information Control Systems Europe, Jisc, Johnson Matthey, Rolls-Royce, Sitryx, University of Edinburgh, University of Salford, University of York, Virgin Media 02.

# 5. About the APPG

The All-Party Parliamentary Group (APPG) on Diversity and Inclusion in Science, Technology, Engineering, and Mathematics (STEM) was established in 2018 and aims to promote the inclusion and progression of people from diverse backgrounds in STEM, and to encourage government, parliamentarians, academics, businesses, and other stakeholders to work towards a STEM sector that is representative of the UK population. We also want to consider and influence changes in policy that will lead to this outcome.

The British Science Association acts as Secretariat for the Group as part of its work on equality, diversity, and inclusion across the science and science engagement sectors.

The Group is made up of Members of Parliament and the House of Lords and is a focus for collaboration with business and other organisations in the STEM sector.

The Officers of the APPG are:

**Chair:** Samantha Niblett MP (Labour)

**Officers:** Baroness Brown of Cambridge (Crossbench)
Lord Lucas (Conservative)
Dr Lauren Sullivan MP (Labour)

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